Chapter 8: Disk Scheduling

ANSWERS

1. Consider one disk with 200 cylinders, numbered 0 to 199. Assume the current position of head is at cylinder 66. The request queue is given as follows:

   55, 32, 6, 99, 58, 71, 86, 153, 11, 179, 42.

Answer questions a) through c) for each of the following disk-scheduling algorithms:

First Come First Served (FCFS)
Shortest Seek Time First (SSTF)
SCAN – in the direction of increasing track number
C-SCAN – in the direction of increasing track number
LOOK – in the direction of increasing track number
C-LOOK – in the direction of increasing track number

a) Show the current position of head based on the disk arm movement to satisfy the above requests, i.e. the sequence of requests serviced.

b) Count the total distance (in cylinders) of the disk arm movement to satisfy the requests.

c) Decide the best algorithm for the given case above. Why?

<u>Answers</u>:

First Come First Served (FCFS)
- a) 66, 55, 32, 6, 99, 58, 71, 86, 153, 11, 179, 42
- b) 11 + 23 + 26 + 93 + 41 + 13 + 15 + 67 + 142 + 168 + 137 = 736

Shortest Seek Time First (SSTF)
- a) 66, 71, 58, 55, 42, 32, 11, 6, 86, 99, 153, 179
- b) 5 + 13 + 3 + 13 + 10 + 21 + 5 + 80 + 13 + 54 + 26 = 243

SCAN
- a) 66, 71, 86, 99, 153, 179, 199, 58, 55, 42, 32, 11, 6
- b) 5 + 15 + 13 + 54 + 26 + 20 + 141 + 3 + 13 + 10 + 21 + 5 = 326

LOOK
- a) 66, 71, 86, 99, 153, 179, 58, 55, 42, 32, 11, 6
- b) 5 + 15 + 13 + 54 + 26 + 121 + 3 + 13 + 10 + 21 + 5 = 286

C-SCAN
- a) 66, 71, 86, 99, 153, 179, 199, 0, 6, 11, 32, 42, 55, 58
- b) 5 + 15 + 13 + 54 + 26 + 20 + 199 + 6 + 5 + 21 + 10 + 13 + 3 = 390

C-LOOK
- a) 66, 71, 86, 99, 153, 179, 6, 11, 32, 42, 55, 58
- b) 5 + 15 + 13 + 54 + 26 + 173 + 5 + 21 + 10 + 13 + 3 = 338

c) Best algorithm is SSTF – least head/disk arm movement.

2. Consider one disk with 2000 cylinders, numbered 0 to 1999. Assume the current position of head is at cylinder 143. The request queue is given as follows:

Pertimbangkan satu cakera dengan 2000 silinder, bernombor 0 hingga 1999. Andaikan kedudukan semasa kepala adalah pada silinder 143. Giliran permintaan diberi seperti berikut:

103, 80, 1400, 813, 1714, 748, 1409, 1100, 1666, 90

For each of the following disk-scheduling algorithms,
Bagi setiap algorithma penjadualan cakera berikut,

i. Shortest Seek Time First (SSTF)

ii. LOOK (variation of SCAN) – in the direction of increasing track number
(dalam arah nombor trek menaik)

iii. C-LOOK (variation of C-SCAN) – in the direction of increasing track number

a) show the current position of head based on the disk arm movement to satisfy the above requests, i.e. the sequence of requests serviced.
tunjukkan kedudukan semasa kepala berdasarkan pergerakan lengan cakera untuk memenuhi permintaan di atas, iaitu jujukan permintaan yang dipenuhi.

b) count the total distance (in cylinders) of the disk arm movement to satisfy the requests.
kirakan jumlah jarak (dalam silinder) bagi pergerakan lengan cakera untuk memenuhi permintaan tersebut.

SSTF	
Track accessed	Distance
103	143 - 103 = 40
90	103 - 90 = 13
80	90 - 80 = 10
748	748 - 80 = 668
813	813 - 748 = 65
1100	1100 - 813 = 287
1400	1400 - 1100 = 300
1409	1409 - 1400 = 9
1666	1666 - 1409 = 257
1714	1714 - 1666 = 48
Total distance	= 1697

LOOK	
Track accessed	Distance
748	748 - 143 = 605
813	813 - 748 = 65
1100	1100 - 813 = 287
1400	1400 - 1100 = 300
1409	1409 - 1400 = 9
1666	1666 - 1409 = 257
1714	1714 - 1666 = 48
103	1714 - 103 = 1611
90	103 - 90 = 13
80	90 - 80 = 10
Total Distance	= 3205

C-LOOK	
Track accessed	Distance
748	748 - 143 = 605
813	813 - 748 = 65
1100	1100 - 813 = 287
1400	1400 - 1100 = 300
1409	1409 - 1400 = 9
1666	1666 - 1409 = 257
1714	1714 - 1666 = 48
80	1714 - 80 = 1634
90	90 - 80 = 10
103	103 - 90 = 13
Total distance	= 3228

3. Access time has two major components: seek time and rotational latency/delay. Explain each of them.

<u>Seek time</u> is the time for the disk are to move the heads to the cylinder containing the desired sector.

<u>Rotational latency</u> is the additional time waiting for the disk to rotate the desired sector to the disk head.